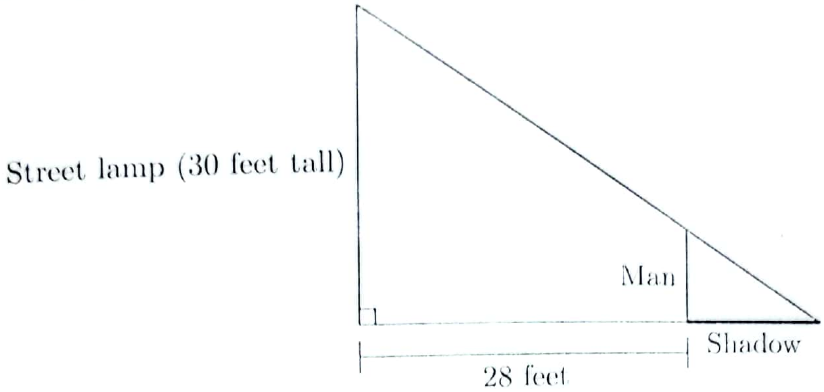
Q1

A man, who is 28 feet away from a 30 foot tall street lamp, is sinking into quicksand. (*See diagram below.*) At the moment when 6 feet of him are above the ground, his height above the ground is shrinking at a rate of 2 feet/second.



Throughout this problem, remember to show your work clearly, and include units in your answers.

a. How long will the man's shadow (shown in bold in the diagram above) be at the moment when 6 feet of him are above the ground?

Solution: Let s be the length of the shadow. Noticing that the larger and smaller triangles in the picture are similar triangles, we have

$$\begin{aligned}\frac{30}{28+s} &= \frac{6}{s}\\ 30s &= 168 + 6s\\ 24s &= 168\\ s &= 7.\end{aligned}$$

So the length of the shadow is 7 feet at that moment.

Answer: 7 feet

b. At what rate is the length of the man's shadow changing at the moment 6 feet of him are above the ground? Is his shadow growing or shrinking at that moment?

Solution: Let h be the height of the man above the ground, and let s be the length of his shadow. Using similar triangles as above, we have $\frac{30}{28+s} = \frac{h}{s}$ so $30s = 28h + hs$.

Taking derivatives with respect to time t, we find $30\frac{ds}{dt} = 28\frac{dh}{dt} + h\frac{ds}{dt} + s\frac{dh}{dt}$.

So at the moment when $h = 6$, we have

$$\begin{aligned}30\left.\frac{ds}{dt}\right|_{h=6} &= 28(-2) + 6\left.\frac{ds}{dt}\right|_{h=6} + 7(-2)\\ 24\left.\frac{ds}{dt}\right|_{h=6} &= -70\\ \left.\frac{ds}{dt}\right|_{h=6} &= \frac{-70}{24} = -\frac{35}{12} \approx -2.917\end{aligned}$$

So at that moment, the shadow is shrinking at a rate of about 2.917 feet/second.

Answer: The man's shadow is (circle one) GROWING $\boxed{\text{SHRINKING}}$

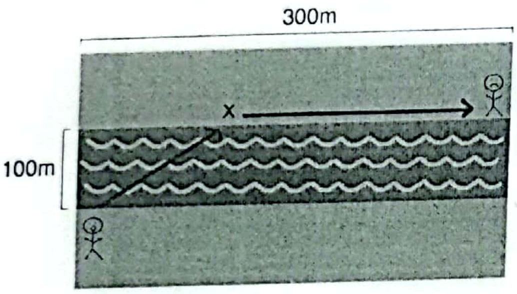
at a rate of $\frac{35}{12}$ **(about 2.917) feet/second**.

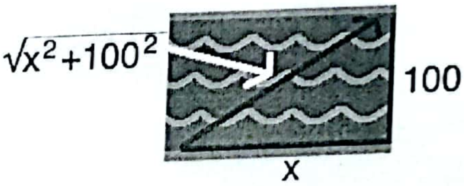
Q2

You are standing on the bank of a river that is 100m wide, and see someone needing help 300m up the opposite shore. You can swim at 3m/s and run at 5m/s, and you want to get to the person as quickly as possible. To what point on the opposite shore should you swim, before running the rest of the way?

Solution:

Let x be the distance up along the opposite shore where I get out of the river and start running. Let T be the total time, which is to be minimized.





$T =$ swim time $+$ run time. Recall that time $= \dfrac{\text{distance}}{\text{speed}}$. How far do I swim?

From the picture we see the swim distance is $\sqrt{x^2+100^2}$. Running distance is $300 - x$. Dividing by the speeds for each distance, this gives

$$T = \frac{\sqrt{x^2+100^2}}{3} + \frac{300-x}{5}.$$

The expression for T is already in terms of only one variable.

$$T' = \frac{1}{3}\left(\frac{2x}{2\sqrt{x^2+100^2}}\right) - \frac{1}{5} = \frac{5x - 3\sqrt{x^2+100^2}}{3\sqrt{x^2+100^2}}. \quad T' = 0 \quad \Rightarrow \quad 5x = 3\sqrt{x^2+100^2}$$

$\Rightarrow \quad 25x^2 = 9x^2 + 9(100^2) \quad \Rightarrow \quad 16x^2 = 9(100^2) \quad \Rightarrow \quad 4x = 3(100) \quad \Rightarrow \quad x = 75.$

So $x = 75$ is a critical number; $T'(70) < 0$ and $T'(100) > 0$ so T is decreasing until $x = 75$ and increasing after, which means there is a relative minimum at $x = 75$. Now, x can be any value in the closed interval $[0, 300]$. So we must check to see if one of the endpoints 0 or 300 is the *absolute* minimum.

$$T(0) = \frac{100}{3} + \frac{300}{5} \approx 93.3, \quad T(75) \approx 66.6, \quad T(300) \approx 105.4,$$

so the shortest time (absolute minimum of T) is 66.6 seconds, which occurs when $x = 75$. Thus, I should swim to a point 75m up the opposite shore and then run the rest of the way.